

INSURELINK

A Unified Insurance Operations Platform — Digitizing the Health Insurance Claim Lifecycle by Unifying Insurers, Hospitals, Corporates, Patients, and Admins on One Real-Time Platform

PROBLEM STATEMENT

- Insurance claim operations in Pakistan remain heavily manual, with hospitals relying on phone calls and emails to coordinate with insurers for approvals.
- Corporate HR departments manage employee insurance data through fragmented spreadsheets with no centralized record-keeping.
- No unified digital platform exists connecting all five stakeholders — insurers, hospitals, corporates, patients, and admins — in one integrated system.
- Patients have zero real-time visibility into claim status, approval progress, or remaining coverage balances.
- High turnaround times, frequent communication errors, and complete lack of audit trails plague the current ecosystem.

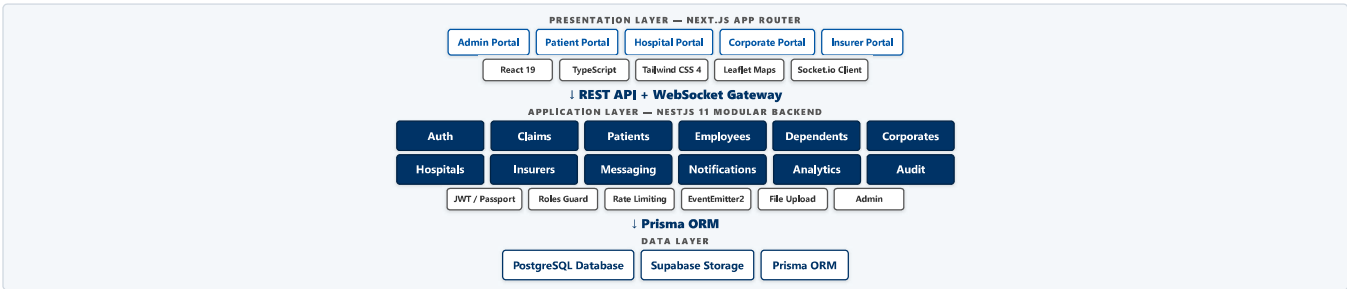
PLATFORM AT A GLANCE

5 Role-Based Portals	50+ REST API Endpoints	15+ Database Entities	38+ Frontend Pages	12 Backend Modules
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CLAIM LIFECYCLE FLOW



SYSTEM ARCHITECTURE



IMPLEMENTATION OVERVIEW

Frontend (38+ Pages)

- 5 Portals:** Admin (12 pages), Insurer (8), Hospital (7), Corporate (5), Patient (6)
- Shared Components:** Layouts, sidebars, notification panels, claim tables, messaging UI
- Real-Time Hooks:** useClaimSocket, useNotifications, useGeolocation, useAuth

Security & Compliance

- JWT Auth:** 15-min access / 7-day refresh tokens
- RBAC:** @RolesGuard with cross-org isolation
- Rate Limiting:** 100/min global, 10/min login
- Audit Trail:** @AuditTable field-level diffs

Backend (50+ REST Endpoints)

- 12 NestJS Modules** with controllers, services, repositories, DTOs
- Socket.IO Gateway** for claim-scoped chat rooms
- Event-Driven Notifications** via EventEmitter2

Data & Validation

- Prisma Schema:** 15+ entities with relations
- Pakistan Validation:** CNIC, mobile, date formats
- Bulk Import:** CSV/Excel → validate → commit → fix invalid

KEY FEATURES & INNOVATIONS

5 Role-Based Portals Dedicated dashboards for Admin, Patient, Hospital, Corporate, and Insurer with tailored workflows.	Real-Time Claims Sync Socket.IO live status updates, notifications, and cross-tab sync across all stakeholder portals.	Smart Hospital Finder Leaflet geolocation map with Haversine sorting, reimbursable filtering, and emergency badges.
Bulk Employee Upload CSV/Excel import with CNIC/mobile validation, duplicate detection, and inline invalid-record correction.	Claims Messaging Real-time chat with attachments, typing indicators, read receipts, and claim-scoped rooms.	Audit & Analytics Field-level diff logging, CSV export, role-based analytics, and coverage utilization tracking.

TESTING & VALIDATION

- API Integration Testing:** All 50+ REST endpoints tested with valid/invalid payloads and edge cases
- Cross-Portal Sync:** Verified real-time claim status propagation between Hospital, Insurer, and Patient portals via Socket.IO
- Bulk Upload Pipeline:** Validated CSV/Excel parsing, Pakistan-specific field validation, duplicate detection, and invalid-record correction flow
- Authentication & Authorization:** Tested JWT token lifecycle, role-based route guards, rate limiting thresholds, and deactivated user blocking
- Audit Trail Integrity:** Confirmed field-level diffs captured for all CREATE, UPDATE, and DELETE operations across modules
- Geolocation Accuracy:** Verified Haversine distance calculations and Leaflet map rendering for Karachi hospital network

TECHNOLOGY STACK



**NATIONAL UNIVERSITY OF COMPUTER
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Final Year Project — Open House 2026

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